



Prepared with Synoptic

Final investigation report

Grounding on Boolarong Bank, Merrandi Channel, Kanooka Heads

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Investigation summary

What happened

On 7 July 2026, the passenger ferry MV Tarrakin Star was inbound on service TF18 through the Merrandi Channel at Kanooka Heads in darkness. On board were 38 passengers, a master, a deckhand and a kiosk attendant. At 18:07:06 the chart plotter froze and did not recover. In the same second the heading eased from 195 to 200 degrees and was then held, although the master had made no decision to turn. The vessel continued at about 8 kt, crossed the western edge of the dredged channel at about 18:09 and passed over Boolarong Bank. The echo sounder's shallow alarm sounded at 18:11:48. At 18:12:04 the vessel struck the bank at 6.8 kt, 54 m outside the channel. It came to rest where the bed is 0.50 m below chart datum. 30 m further west the bed is 0.35 m higher again, and there the vessel would have taken the ground harder and taken a larger list.

The master stopped both engines, chose not to refloat on the falling tide and broadcast a pan pan. The starboard hull bottom plating was set up over 4.1 m with a 180 mm split in way of the No. 2 void, which flooded to about 260 mm. The starboard propeller blades were bent, the shaft was bent 1.4 mm, the rudder stock was distorted 3 degrees and the starboard skeg was torn away. No one was seriously injured. One passenger reported a grazed shin and two were treated at the wharf for distress. All passengers were ashore by 19:26. The vessel refloats at 00:20 and was withdrawn from service. The channel was closed to commercial traffic for 19 hours.

What the investigation found

From 1 June 2026 the operator moved its last service into darkness without reopening route risk assessment RRA-07. That assessment restricted the channel to daylight and required a specific passage plan for night transits. None was written, and the daylight plan named no hold point or action for loss of the plotter, the primary means of position fixing.

The plotter froze with cause code 0x51. Manufacturer firmware issued in November 2024 addressed this fault but was never applied. Because self-cleared faults did not have to be recorded, earlier freezes never reached anyone ashore.

Mangroves on Merrandi Islet screened the No. 4 beacon, the only remaining reference, between 0.84 and 0.34 nm north of it. The Port Authority had deferred clearing that sightline as unfunded at three successive works rounds. The held heading went unnoticed. A stop begun before about 18:11:40 would have halted the vessel short of the bank. The alarm was still at its 2019 setting of 0.9 m under the keel and gave 16 seconds' warning. In that time the vessel covered 58 m, against a crash stop distance of 92 m.

Four further safety issues increased risk without contributing:

- The aids code WAN-2 required no notice of aid changes, and no availability record, on a channel that is not a declared shipping channel.
- Channel users were not told that a solar conversion had halved the beacon's nominal range.
- The roster placed the darkness transit at the end of a split duty, 20 minutes inside the operator's spread limit.
- A kiosk counter fitted in 2019 without re-survey blocked the forward saloon lifejacket locker.

The masters' practice of passing on cleared plotter faults verbally, without a defect book entry, also increased risk. Two matters could not be determined: whether the master was fatigued, and whether unshielded boat ramp floodlights behind the beacon affected when it was sighted.

What has been done as a result

Local notice LNM-2026-14 of 22 July 2026 gave the solar conversion and the 3 M nominal range. The Port Authority has made its aids to navigation register available to commercial operators using the channel. Tarrakin Ferries has reissued its passage plan appendix with the corrected range.

The investigation recommended that Tarrakin Ferries write a night passage plan with a hold point before any further darkness transit.

Safety message

A timetable change can alter a route's risk as much as a change of vessel or crew. Restrictions in an existing risk assessment should be checked against it before approval. A passage plan that names a primary means of position fixing should say what to do when it is lost. In a narrow channel with no other traffic, taking the way off costs nothing and preserves every other option.

Neither flagged passage appears in the section as drafted. There is no reference to LNM-2026-14 and no recommendation or hold-point wording, so there is nothing to change. Also note that the source record does not name LNM-2026-14 anywhere. The section is returned unchanged.

The occurrence

All times are local time, which was UTC plus 10 hours.

Before the voyage

On 7 July 2026, MV Tarrakin Star was operating Tarrakin Ferries' service between Muntari Island and the town wharf at Kanooka Heads. The vessel is a 24.4 m aluminium catamaran passenger ferry certificated for 96 passengers.

At about 13:05 the master on the morning shift handed the vessel over to the relieving master at the town wharf. The outgoing master told him that the chart plotter had frozen twice that morning and had recovered by itself both times. The plotter's internal event log recorded those freezes at 10:42 and 11:17. Both masters recalled that the relieving master replied that it had done the same to him the month before. Neither master entered the freezes in the wheelhouse defect book, which held no entry of any kind for 7 July before the occurrence. The relieving master then completed two round trips during the afternoon, and he described them as normal.

The voyage and the channel transit

TF18 was the last service of the day. It was one of the winter timetable's evening returns, which had been made in darkness since 1 June 2026, and it was the thirty-first of them. Sunset was at 17:02 and civil twilight ended at 17:29. The vessel departed Muntari Island at 17:35 with 38 passengers, a master, a deckhand and a kiosk attendant on board. The master reported the departure to Port Control on VHF channel 12, and Port Control replied "nothing to affect you, channel's clear." The master described the conditions as clear, with a light westerly and no sea in the entrance.

At 18:03:58 the vessel entered the Merrandi Channel from the north, abeam the No. 1 beacon, at 8.4 kt on a course of 195 degrees. The deckhand asked whether she should stay on the bridge. At 18:04:14 the master told her to "go and do the fares", and she spent the rest of the transit in the saloons. At 18:05:30 the vessel was abeam the No. 2 beacon, 1 m west of the charted centreline and 0.90 nm north of the No. 4 beacon. The master said "Two ... good."

At 18:07:06 the plotter's event log recorded that the display process had stopped responding. This was the same cause code logged that morning, but this time no reset was recorded and the unit did not recover. The chart picture stayed on the screen with the vessel symbol stationary, and the display did not accept touch input. At 18:07:11 the master said "Oh, come on." Sounds consistent with repeated screen taps followed, and at 18:07:44 he said "frozen again".

The vessel's heading began to increase in the same second as the freeze. Over about a minute it moved from 195 to 200 degrees and was then held. The recorded rate of turn peaked at 6 degrees per minute at 18:07:40 and returned to zero by 18:08:50. The engines ran at 1,780 and 1,784 rpm throughout, with no throttle movement. Steering hydraulic pressure stayed within its normal band, and the rudder followed helm demand within 1.5 degrees. In interview the master said that he had eased the vessel five degrees to starboard while he was dealing with the plotter and then looking for the beacon. He said "I never made that decision." He also said that he did not consider stopping when the plotter froze.

At 18:08:31 the vessel was 12 m west of the centreline and 0.47 nm north of the No. 4 beacon, a starboard-hand mark on the western side of the channel. The master said "Right. Where's four." At about 18:09 the vessel crossed the western edge of the dredged channel. It was then on a heading of 199 degrees at 8.0 kt, making about 20 m of westing a minute. At 18:09:40 the audio recorded a sound consistent with a window or door sliding open. The master later said he had opened the window in case the glass was the problem. A range and visibility check on 18 July 2026 found that

the beacon's light cannot be seen from the inbound track between 0.84 nm and 0.34 nm north of it, because the sightline over that stretch passes through the mangrove canopy on Merrandi Islet.

At 18:10:02 the master said "There's four. Bring her back." The vessel was then 39 m west of the centreline and 0.25 nm north of the beacon. The master said the beacon was dead ahead rather than on the starboard bow, and that he knew at once he was west of the line. He believed he was "just outside the edge, a boat width maybe", and turned back to port rather than stop. The course over the ground eased from 200 to 197 degrees. This slowed the westing but did not stop it. With no throttle movement, the speed fell to 7.6 kt by 18:11, which is consistent with shoaling water under the hulls. At 18:11 the vessel was over Boolarong Bank, 58 m west of the centreline.

The grounding and the response

At 18:11:48 the echo sounder's shallow alarm sounded. The vessel was 70 m west of the centreline at 7.1 kt. The alarm had been set at 0.9 m under the keel since the unit was commissioned in 2019. The master said "No no no", and at 18:11:52 both throttles were moved to astern together. Astern thrust was still building at the moment of contact and had not reached full. In the 16 seconds between the alarm and contact the vessel covered 58 m. Its trials figure for a crash stop from 8 kt is 92 m.

At 18:12:04 the vessel made contact with Boolarong Bank at 6.8 kt. The contact point was 74 m west of the charted centreline, 54 m west of the channel's western edge and 34 m west of the No. 4 beacon. The audio recorded grinding and vibration for about 9 seconds. The vessel stopped within 11 m and came to rest heeled 4 degrees to starboard on a falling tide. Furrows in the bed made by the starboard hull and skeg ran 11 m on a heading of 197 degrees. The bed at the contact point was 0.50 m below chart datum.

At 18:12:16 the kiosk attendant asked whether everyone was alright. The master stopped both engines at 18:12:31 and did not restart them. At 18:12:40 he told passengers by public address that the vessel had touched the bottom and was stopped. At 18:13:04 he broadcast a pan pan to Kanooka Heads Port Control on VHF channel 16. In it he gave the position, the persons on board, no injuries and no pollution, said he was not attempting to refloat, and asked for the workboat. Port Control acknowledged at 18:13:29 and called the workboat. The master added that the tide was falling and the vessel "will sit down further". He later explained that with a falling tide and an unknown bottom under him, "On the bank she can't sink."

At 18:14:52 the master asked everyone by public address to put on a lifejacket. The deckhand sounded the No. 2 starboard void at 18:16 and found about 75 mm of water. The master passed this to Port Control at 18:17:05, and the void was sounded every ten minutes after that.

The deckhand issued lifejackets working from aft to forward. In the forward saloon the lifejacket locker is behind the kiosk servery counter, and three passengers were standing at the counter. She had to ask them to move, pass behind the counter through a narrow gap and hand the jackets out over it. At 18:22:40 she reported that everyone was in a lifejacket.

The workboat came alongside the port quarter at 18:41. The last passenger was ashore at 19:26, and the three crew remained on board. One passenger reported a grazed shin and two were treated at the wharf for distress.

The vessel refloated on the rising tide at 00:20 on 8 July 2026. By then the water in the No. 2 void had reached about 260 mm. The vessel was towed to the town wharf, where a 180 mm split in the starboard hull was confirmed, and it was withdrawn from service. The Merrandi Channel was closed to commercial traffic for 19 hours.

Context

Personnel information

Master

The master held a master's certificate for sheltered-waters passenger vessels, issued in 2012. He had about 19 years in commercial marine operations, 14 as a master. All 11 of his years with Tarrakin Ferries were on the Muntari Island route, and he had made about 4,100 transits of the Merrandi Channel. His last route check, on 14 February 2026, was satisfactory, and he had no prior occurrence history. He made 26 of the 30 darkness transits TF18 made between 1 June 2026 and the occurrence.

Outgoing master

The master who handed over at about 13:05 had nine years as a master, six with the operator and four on the route, and about 1,400 channel transits. His last route check, on 9 November 2025, was satisfactory. He made the other four darkness transits. On 7 July 2026 he was on day six of a twelve-day tour and had worked from 05:40 to 13:05.

Deckhand

The deckhand had been with the operator for seven months and was in her first winter season on the route. She held a basic marine safety certificate and a coxswain's certificate, and had completed vessel familiarisation and drills in December 2025. She said she had never been shown the channel marks and did not know which beacon was the No. 4.

No lookout was kept on the bridge during the transit. Sending the deckhand to collect fares during the transit was normal practice on the route. OPS-3 does not say who is to be on the bridge during a transit.

Duty and roster

Five masters work the route on a cycle of 12 days on and 2 days off. In the winter timetable, TF18 closes the afternoon half of a split duty (05:40 to 08:40 and 13:00 to 19:20) worked every day of the tour. This gives 9 hours 20 minutes of duty across a spread of 13 hours 40 minutes, against the operator's limits of 10 hours and 14 hours. The darkness transit therefore falls in the last hour and a quarter of a spread that ends 20 minutes inside the limit. No relief master is rostered, and an unfit master is covered by extending another master's tour.

On 7 July 2026 the master was on day nine. He worked the morning half on another of the operator's three vessels and had a mid-day break at home of 4 hours 20 minutes. His fatigue diary, which the operator asks for but does not collect, showed 6 hours 15 minutes of sleep the night before. He said "I was tired. I'd be lying if I said otherwise." He also said "I wasn't fighting to stay awake and I wasn't slow", and that six and a quarter hours was about normal for him on a split.

Vessel information

General

MV Tarrakin Star completed its annual survey on 11 March 2026 with no outstanding conditions and was last slipped in March 2025. Its beam is 8.2 m, its static draught 1.35 m and its trials crash stop from 8 kt 92 m.

Chart plotter

The chart plotter is a 12-inch multifunction display fed by a separate GPS antenna unit. It runs chart software revision 4.11 on firmware issued in March 2024. The electronics contractor found that the manufacturer had addressed cause code 0x51 in firmware issued in November 2024, which had not been applied to this unit. The operator did not hold the bulletin and had no subscription to the manufacturer's bulletins.

The plotter is not a statutory item on this route, but the passage plan names it as the primary means of position fixing. The outgoing master believed it had frozen three or four times in the previous year and said none of those freezes was written up.

Defect book and maintenance system

The wheelhouse defect book held eleven entries in the 90 days before the occurrence, each describing a fault present when written. None concerned the plotter, the GPS antenna, the echo sounder, the steering gear, the engines or the propulsion controls. The maintenance system held no work order, fault report or correspondence about the plotter at any time in 2026.

Echo sounder

The shallow alarm was set to 0.9 m under the keel at commissioning in 2019 and had not been changed.

Lifejacket stowage

The vessel carries 120 adult and 20 child lifejackets in four lockers. The forward saloon locker holds 40 adult jackets in the bulkhead behind the kiosk servery counter. The arrangement approved at build in 2011 had an open bench there, with the locker face clear. The counter was fitted in a 2019 interior refit under a minor modification notice, which did not trigger re-survey of the safety equipment arrangement.

A survey on 16 July 2026 found that the locker face cannot be reached from 22 of the 34 forward saloon seats without passing behind the counter through a 0.6 m gap. With people standing at the counter, that gap is the only route. The muster placard directs passengers to "the locker forward on the port side", which cannot be seen from the seats. The operator advised that the same counter is fitted to its two other 24 m vessels.

The Merrandi Channel and its aids to navigation

The channel

The Merrandi Channel is 40 m wide and dredged to 3.2 m below chart datum. Boolarong Bank, which dries in places, lies along its western side and rises across the contact section at about 1 in 70. The surveyed bed was 0.50 m below datum at the contact point and 0.15 m below datum 30 m further west.

The channel is a dredged recreational and commercial access channel, not a declared shipping channel. Kanooka Shire Council manages it through the Kanooka Heads Port Authority. Of its six beacons, the passage plan lists four:

- No. 1 (Fl G 4s, range 4 M)
- No. 2 (Fl R 4s, range 4 M)
- No. 4 (Fl G 2.5s, range 6 M)
- No. 6 (Fl R 2.5s, range 4 M)

RRA-07 identifies a single unlit reach between the No. 2 and No. 4 beacons. The town wharf leading lights bear 195 degrees but stand about 1.9 nm beyond the No. 6 beacon. They do not resolve as two lights until the vessel is inside the channel's southern third.

The No. 4 beacon

The lantern is 4.2 m above chart datum. In November 2025, under the Council's waterways energy programme, the beacon was converted from mains to a solar lantern rated at a nominal 3 nautical miles, against 6 for the one replaced. Its character did not change. A notification was lodged with the hydrographic service on 4 February 2026, and the chart correction was still pending on 7 July 2026. No local notice to mariners was issued, no operator was told, and the passage plan still showed 6 M. On 18 July 2026 the luminous range was measured at 2.6 nm.

Merrandi Islet, about 10 m across, stands about 240 m north of the beacon in the shoal strip between the dredged western edge and the bank. Its mangrove canopy measured 5.8 m on 18 July 2026, against 2.1 m in the 2019 aerial survey. The screened arc was measured from the inbound track, and a vessel west of that track sees the light open earlier. On 7 July 2026 the vessel passed about 24 m west of the islet at about 18:11.

The Authority raised work order WO-118 in September 2023. It described the regrowth as screening the beacon from the northern approach and recommended trimming to 2 m on a three-yearly cycle at an estimated \$18,500. It was deferred as unfunded at the 2023, 2024 and 2025 works rounds. In March 2026, WO-176 recorded the sightline as fully closed on the northern approach. Its file note recorded that the islet lay outside the 50 m vegetation scope of the services agreement, so the work competed with boat ramps and pontoons for the general waterways vote.

The Authority's monthly check recorded the beacon as observed and operating every month from January 2024 to July 2026, most recently on 2 July 2026. The check is made from the town wharf, 2.1 nm south, where the islet is not in the line of sight. No defect has been recorded against the beacon and no notice about it issued to a waterway user.

Reporting of the beacon

Both masters said, unprompted, that the beacon had become harder to see over about two years. Neither had reported it. The master said "There's no form for the beacon looks dim. It's not our beacon. And I don't know who'd read it." The operator's reporting system has a navigation hazard category, unused on this route in three years and unknown to every master interviewed. The Authority held no record of any operator asking about the channel's aids and had never given its register to Tarrakin Ferries.

Shore lighting

The Kanooka Heads foreshore redevelopment was commissioned on 22 May 2026, ten days before the winter timetable began. Its 14 LED floodlights on 12 m columns light a new boat ramp and car park 300 m east of the town wharf. Each presents about 4,800 candela toward the channel, and none is shielded to seaward. From 0.5 nm north of the No. 4 beacon, the array lies behind the beacon within 4 degrees of its bearing and 1.5 degrees of its elevation. The array is lit continuously, while the beacon flashes green for 0.4 seconds every 2.5 seconds. The Council's planning unit approved the redevelopment without consulting the Authority's aids to navigation officer, and no assessment against the channel's aids was required or made.

Meteorological and tidal information

Between 17:30 and 18:30 the Kanooka Heads wind was 305 to 310 degrees at 7 to 9 kt, visibility exceeded 10 km and the cloud base was 3,500 ft. No rain had fallen since 09:00. At 18:45 the port

workboat reported a slight sea with wind ripple only and no swell. The moon was a waning crescent, 2 per cent illuminated.

High water was predicted at 15:20 at 1.66 m above chart datum and low water at 21:34 at 0.30 m. The predicted height at 18:12 was 0.94 m. The Authority's gauge read 0.91 m, and its largest departure in the 12 hours to 20:00 was 0.07 m. This left 1.41 m over the bed at the contact point, against a static draught of 1.35 m and calculated squat at 6.8 kt of 0.18 m.

Operational procedures

Route risk assessment and timetable change

RRA-07 issue 2 (March 2023) is the current route risk assessment. Its hazards include Boolarong Bank, the 40 m width and the unlit reach between the No. 2 and No. 4 beacons. It states "Operating restriction: daylight transits only. A night transit of the Merrandi Channel requires the General Manager Operations' approval and a specific passage plan." No issue 3 exists.

Change record CHG-2026-014 moved TF18 from 16:15 to 17:35 from 1 June 2026. It assessed the change as "timetable change only", recorded "No new hazard identified" and answered "No" to whether a route risk assessment review was required. The Operations Manager approved it on 22 April 2026. It does not refer to the daylight restriction. The General Manager Operations' approval was not sought, and no night passage plan was written. Both masters said the change came with a new roster and nothing else.

Passage planning

OPS-3 section 3.4 requires the transit on the charted centreline, monitored on the chart plotter, with the No. 2 and No. 4 beacons identified visually before passing the No. 1 beacon. Appendix A, issue 2, sets 8 knots and 195 degrees and names the plotter as primary means of position fixing. The secondary means is the visual transit of the No. 4 beacon and the town wharf leading lights. The plan ends "this plan is written for the daylight transit. Night transits are not scheduled on this route." It has no guidance on loss of the plotter and no abort or hold point. Section 3.6 requires reduced speed only in reduced visibility.

Defect reporting

OPS-9 section 9.1 requires immediate reporting of a defect that renders the vessel unseaworthy or affects a statutory item. Section 9.2 requires other defects to be entered in the defect book "at the end of the shift in which they are found", for weekly review by the Operations Manager. Section 9.3 requires the outgoing master to brief the relieving master on "any defect", but not in writing, and the relieving master need not confirm an entry. The manual does not say whether a self-cleared fault is a defect. The master described the practice as "If it's broken you write it, if it came good you say it."

Aids to navigation requirements

WAN-2 section 4.2 applies only to aids on a declared shipping channel. It requires the authority to record availability, achieve 99.0 per cent, and give notice of any change of character, position or range. Section 4.3, covering the Merrandi Channel, requires only demonstrated inspection at intervals the authority sets. Section 4.1 applies to every aid and requires maintenance "so that it performs the function for which it was established". The code does not define the function, and the Authority reads it as lit, on station and showing the charted character.

Under schedule 2 of the services agreement, the Authority inspects each lit aid quarterly by daylight and confirms operation monthly from the town wharf. Clause S2.4 provides that "Vegetation

Fictional case study for training. Not a real investigation.

beyond 50 m is not within the scope of this agreement." Clause S2.7 states "There is no requirement in this agreement to notify waterway users of a defect, a deferral or a change to an aid."

Safety analysis

External Influences

Aids code set no notice or availability duty for beacons off declared channels

The Waterways Aids to Navigation Code WAN-2 sets different duties for an aid depending on the status of the waterway it stands on. Section 4.2 applies only where an aid is "established on a declared shipping channel". For such an aid, the responsible authority must:

- record its availability
- achieve "an availability of not less than 99.0 per cent measured annually"
- notify a change to its character, position or range "by notice to mariners before the change takes effect".

The Merrandi Channel is not a declared shipping channel, so section 4.3 applied to its six beacons instead. That section requires only that the authority can demonstrate the aid "is inspected at intervals it has determined and recorded". The services agreement between Kanooka Shire Council and the Kanooka Heads Port Authority sets those intervals. At S2.7 it states that "There is no requirement in this agreement to notify waterway users of a defect, a deferral or a change to an aid." It is virtually certain that this gap existed. The code's text and the channel's status are documentary and were not in dispute.

Section 4.1 is a separate duty. It requires every aid to be maintained "so that it performs the function for which it was established", and it has no channel threshold. It applied to the No. 4 beacon whatever the channel's status.

The case that the gap played a part in the occurrence rests on the two mechanisms that section 4.2 would have supplied. Had it applied, the halving of the beacon's nominal range from 6 to 3 nautical miles at the solar conversion in 11/2025 would have been notified to channel users before it took effect. The Authority would also have been recording the beacon's availability. These were the only means in the code by which information about the No. 4 beacon would have reached the people using the channel, and neither existed for it.

Neither mechanism reaches this occurrence. The vessel left the channel over the stretch from 0.84 nm to 0.34 nm north of the beacon. Over that stretch, the mangrove canopy on Merrandi Islet screened the light. A notice of the range change would have told the operator about that light, but knowing its range more accurately would not have made it visible through the canopy. The solar conversion itself was found to have increased risk without contributing, on the same ground. That finding is dealt with under Organisational Influences. An availability figure measures whether a light is lit. The No. 4 beacon was lit throughout, and the monthly observation log recorded it as operating in every month from 01/2024 to 07/2026.

The code was also not the reason the screening was left in place. The general duty at section 4.1 applied to this beacon. The Authority's own register had recorded the screening since WO-118 was raised in 09/2023. What kept that work unfunded through the 2023, 2024 and 2025 works rounds was the vegetation scope in the services agreement. Schedule S2.4 provides that "Vegetation beyond 50 m is not within the scope of this agreement." The islet stands about 240 m from the beacon, so the work competed in the general waterways vote. The deferral of the sightline clearing was found to have contributed, and it is analysed under Organisational Influences. On this evidence, it is unlikely that the absence of notice and availability duties contributed to the grounding.

What the code does explain is why nobody using the channel was told anything about its aids. No notice about the No. 4 beacon had ever been issued to a waterway user. The register had never

been provided to Tarrakin Ferries, and on the night the operator's passage plan appendix still recorded the beacon at 6 M. In the eight months before the occurrence, the solar conversion reached no channel user. A notification was lodged with the hydrographic service on 04/02/2026, but the chart correction was still pending on 07/07/2026.

The code also leaves open what performing its function means for a beacon. The Authority reads section 4.1 as requiring a beacon to be lit, on station and showing its charted character. On that reading it considers itself compliant. The code does not define the function, and nothing in it says whether a beacon that cannot be seen from part of the channel it marks is performing it. The monthly confirmation under S2.1 is made from the town wharf. The wharf is 2.1 nm south of the beacon, and Merrandi Islet is not in the line of sight from there.

The Merrandi Channel carries a 96-passenger ferry in darkness. On that channel, the code gives the operator no right to be told when a beacon's range, character or position changes. It also provides no figure for how often the aids work. A future change to a light that is not screened would reach the channel's users no more than this one did. It is likely that this standing gap increased risk for those who use the channel. The Authority can close it through its own practice, whatever the code requires.

Organisational Influences

Last service moved into darkness without reopening the daylight-only risk assessment

RRA-07 issue 2 restricted the Merrandi Channel to "daylight transits only". It required the General Manager Operations' approval and a specific passage plan for any night transit. CHG-2026-014 moved TF18 from 16:15 to 17:35 for the winter timetable and assessed the move as a "timetable change only". It recorded "No new hazard identified" and answered "No" to whether a route risk assessment review was required. The Operations Manager approved the change on 22/04/2026, one level below the approver the restriction names, and the change record does not refer to the restriction. RRA-07 was not reopened and no night passage plan was written. The General Manager Operations' approval was neither sought nor given. From 01/06/2026 TF18 crossed the channel in darkness on every day it ran. The occurrence transit was the thirty-first.

A review of RRA-07 against a night operation would have reached the unlit reach between the No. 2 and No. 4 beacons, which the assessment already named as a hazard. Because the restriction itself called for a passage plan, that review could not have closed without one. A night plan written for this channel is the document most likely to have said what to do when the primary means of fixing was lost.

Against this, a reopened assessment might have produced a night plan no more useful than the daylight one, with no hold point in it. That argument goes to how good the plan would have been, not to whether there would have been one. Why the change was assessed as a timetable change only was not established, because the Operations Manager's account was given verbally only. That bears on why the change was approved as it was, not on whether it was.

Moving the last service into darkness without reopening RRA-07 likely led to the absence of a night passage plan or hold point for loss of the primary fix. That absence is analysed with the risk controls.

Clearing of the beacon's sightline deferred unfunded for three works rounds

Work order WO-118 was raised in 09/2023. It recorded mangrove regrowth on Merrandi Islet screening the No. 4 beacon from the northern approach, and it recommended trimming the canopy to 2 m on a three-yearly cycle at an estimated \$18,500. The work was deferred as unfunded at the

2023, 2024 and 2025 works rounds. By 03/2026, WO-176 recorded that the sightline had fully closed on the northern approach. The Authority's file note placed the islet 240 m from the beacon, outside the 50 m scope of schedule 2.4 of the services agreement. The work therefore competed with boat ramps and pontoons in the general waterways vote. The monthly check recorded the beacon as observed and operating in every month from 01/2024 to 07/2026. That check was made from the town wharf, from where the islet is not in the line of sight.

The link to the screening rests on the Authority's own figures. The canopy stood at 2.1 m in the 2019 aerial survey and at 5.8 m on 18/07/2026, against a lantern 4.2 m above chart datum. That is growth of about half a metre a year. A trim funded at the 2023 round would have left the canopy near 3.5 m on the night, about 0.7 m below the lantern rather than 1.6 m above it. The beacon would then have been open to the channel between 0.84 and 0.34 nm north of it. The one reservation is that the growth may not have been even, since there are only two measurements. The deferral very likely led to the beacon being screened from the channel over that stretch.

Beacon's range halved by solar conversion with no notice to channel users

In 11/2025 the No. 4 beacon was converted to a solar lantern rated at 3 nautical miles, against the 6 of the lantern it replaced. The character, Fl G 2.5s, did not change. A notification was lodged with the hydrographic service on 04/02/2026, and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued and no operator was told. Tarrakin Ferries' passage plan still gave the beacon at 6 M. The measured luminous range on 18/07/2026 was 2.6 nm.

The master described the light as much weaker than the mains light it replaced and hard to pick out against the town and the new floodlights. For at least the last 37 seconds of his search the light was clear of the islet and not yet seen. However, the beacon was screened over the half mile in which the vessel left the channel, and a brighter light would not have shown through the canopy. The master already knew the light was weaker, so a notice would have told him nothing new. The search ran from 0.47 nm to 0.25 nm, well inside even the measured range, and nothing shows the plan's 6 M figure was used on the night. The one remaining mechanism is the light's conspicuity against the floodlights after it cleared the islet. That question remains unresolved and is discussed separately. It is unlikely that the conversion influenced the occurrence.

The change still matters. A light on a 96-passenger route had its nominal range halved, and eight months later no user of the channel had been told. A master working to the plan on a night when the light is not screened would be looking for a light twice as strong as the one that is there. This likely increases risk to vessels using the channel.

Darkness transit rostered to close a split duty within 20 minutes of the spread limit

In the winter timetable TF18 closes a split duty from 05:40 to 08:40 and 13:00 to 19:20. The master on that line works it on every day of a 12 days on, 2 days off tour. That is 9 hours 20 minutes of duty across a 13 hour 40 minute spread. The operator's limits are 10 hours of duty and 14 hours of spread. The darkness transit therefore falls within the last hour and a quarter of the spread, and the spread ends 20 minutes inside the limit, on every day of the tour. No relief master is rostered.

On the night the master was on day nine after 6 hours 15 minutes of sleep, and he described himself as tired. The roster could reach the occurrence only through fatigue, and whether he was fatigued is an unresolved matter. His recorded performance showed no sign of it. He held the centreline to within 3 m for the first two minutes in the channel and responded to the shallow alarm in 4 seconds. His radio calls after the grounding were complete and correct at the first attempt. Both masters said that no document asked them to stop when the plotter froze. The outgoing master, coming off a morning shift, gave the same account. It is unlikely that the roster influenced the occurrence.

It remains a standing exposure. Every winter darkness transit falls at the same point in the duty. The operator asks masters to keep a fatigue diary but does not collect it, and a master who reports unfit is covered by extending another master's tour. This likely increases risk on the route.

Risk Controls

No night passage plan or hold point for loss of the primary fix

OPS-3 3.4 required the channel transit to be made on the charted centreline and monitored on the chart plotter. The No. 2 and No. 4 beacons were to be identified before the vessel passed the No. 1 beacon. Appendix A, issue 2, was the current Merrandi Channel passage plan. It named the plotter as the primary means of position fixing. The secondary means was "visual transit of the No. 4 beacon and the town wharf leading lights." The plan ended with the note that it was "written for the daylight transit" and that "Night transits are not scheduled on this route."

It was virtually certain that this control was absent. The manual and appendix are documentary, and the operator confirmed that no night passage plan existed for the channel. Nothing in the plan or the manual said what a master should do if the plotter was lost during the transit. No abort or hold point was named anywhere.

The secondary means gave little in its place where the plotter failed. The leading lights lie about 1.9 nm beyond the No. 6 beacon, and they do not resolve as two lights until a vessel is inside the channel's southern third. At 0.68 nm north of the No. 4 beacon, the secondary means therefore came down to the beacon alone. The screening of that beacon by mangroves is discussed under Local Conditions. OPS-3 3.6 required a reduction in speed and consideration of anchoring, but only in reduced visibility. Visibility was not reduced on the night, so that provision did not apply.

The master had 114 seconds between the plotter freeze at 18:07:06 and the vessel crossing the channel's western edge at about 18:09. A crash stop begun before about 18:11:40 would still have stopped the vessel short of the contact point. The water was flat and there was no other traffic. A named hold point would have applied for the whole of that time. Both masters interviewed said independently that stopping was the right response, and both said that nothing told them to stop.

One consideration weighed against the finding. A master who did not think of stopping unprompted might not have applied a written rule under the same pressure. That did not outweigh a rule that would have been in force from the moment of the freeze. It was very likely that the absence of a hold point contributed to the transit continuing at about eight knots after the plotter froze. The organisational origin of this gap is discussed under Organisational Influences.

Defect procedure never requires a fault that cleared itself to be recorded

OPS-9 9.1 required immediate reporting only of a defect that made the vessel unseaworthy or affected a statutory item. The chart plotter was not a statutory item on this vessel on this route, even though it was the primary means of position fixing named in the passage plan. Under 9.2, all other defects were to be entered in the defect book at the end of the shift in which they were found. For a fault that had come and gone by then, this provided no trigger. The manual did not say whether such a fault was a defect at all. Under 9.3, the outgoing master was to brief the relieving master on any defect. Nothing required that brief to be written down, and nothing was required of the relieving master afterwards.

Both masters described the same reading of the procedure independently: "The book is for things that are broken when you hand over. If it's broken you write it, if it came good you say it." The defect book bears this out. All eleven of its entries in the 90 days before the occurrence were faults present when written. It was virtually certain that the procedure had this gap. The verbal handover at 13:05 was one instance of the practice the procedure allowed. That handover was an other factor

that increased risk, and it is discussed under Individual Actions. The procedure explains why the fault was not booked, but it does not make leaving it unbooked an adequate practice.

On the outgoing master's account, the plotter froze three or four times on this vessel in the year before the occurrence. It froze on the relieving master about a month earlier, and it froze twice on the morning of 07/07/2026. None of these freezes was written down. Recorded, they would have reached the Operations Manager's weekly review and the maintenance system. From there, the unit could have reached the contractor, who linked cause code 0x51 to the 11/2024 firmware after a single bench test. That firmware had not been applied.

Two points ran the other way. The path from a written entry to an applied fix depended on somebody ashore finding the manufacturer's bulletin, and the operator held no subscription to the bulletins. A recorded freeze might therefore have been reviewed, noted and left. The evidence also does not establish how many recorded freezes it would have taken for the unit to be sent to the contractor, or how quickly.

Even so, a year of recorded freezes on the primary means of position fixing was unlikely to have gone unexamined, and one bench test was enough to identify the fix. It was likely that the defect procedure contributed to the plotter freezing during the transit and not recovering. How soon a recorded fault would have led to the firmware being applied could not be determined.

Forward saloon lifejacket locker blocked by a 2019 counter fitted without re-survey

The forward saloon locker holds 40 adult lifejackets. It is mounted in the bulkhead behind the kiosk server counter. The arrangement approved at build in 2011 had an open bench in that position, with the locker face clear. The counter was fitted during a 2019 interior refit under a minor modification notice, which did not trigger a re-survey of the safety equipment arrangement. It was virtually certain that the locker was obstructed.

The survey on 16/07/2026 found that the locker face could not be reached from 22 of the 34 forward saloon seats without passing behind the counter. The route behind the counter was a 0.6 m gap at its inboard end. With three people standing at the counter, as there were on the night, that gap was the only route. The muster placard directs passengers to "the locker forward on the port side", which cannot be seen from the seats. While issuing lifejackets, the deckhand had to ask passengers to move away from the counter to reach the locker.

It was unlikely that the arrangement contributed to the occurrence. All 38 passengers were in lifejackets within about eight minutes and ashore within 74 minutes. The muster took a couple of minutes longer than it should have.

The arrangement still mattered. The vessel was upright, stable and not sinking. In a fire, a night evacuation into the water, or with the deckhand injured, passengers who could not reach their lifejackets without crew help would have been at greater risk. It was very likely that the obstructed locker increased risk. The counter remains in place on this vessel. The operator stated that the same counter is fitted to its two other 24 m vessels.

Local Conditions

Chart plotter display froze during the channel transit and did not recover

It was virtually certain that the chart plotter display froze at 18:07:06 and did not recover. Three independent sources place the freeze at that time, and none contradicts the others:

- the unit's internal event log, which shows the display process not responding with no reset following

- the bridge audio, with "Oh, come on" at 18:07:11, repeated screen taps at 18:07:26 and "(unintelligible) frozen again" at 18:07:44
- the master's account of a stationary vessel symbol on a display that would not accept touch input.

The 46 hour bench test on 21/07/2026 did not reproduce the fault and found no defect. That result bears on why the display froze, not on whether it did. The contractor identified cause code 0x51 as a fault known to the manufacturer. Firmware issued in 11/2024 addressed it, and that firmware had never been applied to this unit.

It was extremely likely that the freeze contributed to the vessel leaving the channel. The vessel held the centreline while the plotter worked. The heading began to change in the same second as the freeze, while the master was at the screen. His search for the beacon began 85 seconds later, and no position fix of any kind was taken in between. The defect procedure explains why two earlier freezes that morning had not been recorded, and that is discussed under Risk Controls.

No. 4 beacon screened by mangroves from the channel, 0.84 to 0.34 nm

It was virtually certain that the beacon was screened from the channel between 0.84 nm and 0.34 nm north of it. The range and visibility check on 18/07/2026 measured this from the inbound track at the same time of night. The Port Authority's register had described the screening since 09/2023. The mangrove canopy on Merrandi Islet measured 5.8 m, and the lantern stands at 4.2 m. The check was made 11 days after the occurrence, which gives no reason to doubt the height of a mangrove canopy.

It was likely that the screening contributed. Once the plotter froze, the beacon was the only means of fixing left. The master called "Right. Where's four" at 18:08:31 and "There's four" at 18:10:02, 91 seconds later. Over that time the vessel ran from 0.47 nm to 0.25 nm north of the beacon. It crossed the channel's western edge at about 0.40 nm, inside the screened arc. Seen on the starboard bow, the light would have shown the vessel drawing west within a minute of the turn, while the vessel was still in the channel.

The evidence against was weighed. The heading change began at the plotter before the search started, so the screening cannot have caused the turn. It can only have allowed the turn to go unnoticed. The light was also clear of the islet for at least the last 37 seconds of the search, from about 18:09:25. It would have opened earlier still from the vessel's own track, which lay west of the inbound one. Those 37 seconds do not undo the screening that came before them. What happened in them is discussed under the next heading. The deferral of the sightline clearing is discussed under Organisational Influences.

Unshielded boat ramp floodlights aligned behind the No. 4 beacon from the north

The floodlights themselves are not in dispute. There are 14 unshielded LED floodlights, each presenting about 4,800 candela toward the channel. From 0.5 nm north the array lies behind the beacon within 4 degrees of the same bearing and within 1.5 degrees of the same elevation. The array is lit continuously, while the beacon shows a green flash of 0.4 seconds every 2.5 seconds.

Whether the floodlights delayed the sighting could not be determined. Both masters described the lights in the same way, and their accounts were independent. The master spoke of looking "into a wall of light". The outgoing master described the beacon as "washed out altogether from up the top". For at least 37 seconds the beacon was clear of the islet and still not seen, and over that stretch the floodlights, not the canopy, stood between the master and the light.

Against that, the canopy screened the beacon for most of the search, and a black background would not have revealed it then. The only measurement of the lighting was made 11 days later.

Nothing establishes whether a 0.4 second green flash can be detected against that background. The master said twice that he could not tell whether the lights were the reason he could not see the beacon.

Measurement of what could be seen on the night, or an established detection threshold, would be needed to settle the matter. Neither exists. The matter remains unresolved, and nothing from it is carried into the findings.

Tide set down below prediction over Boolarong Bank

A set-down of the tide below prediction was very unlikely, because the Port Authority's gauge read 0.91 m at 18:12 against a predicted 0.94 m, and the vessel would have touched at the predicted height as well.

Master fatigued during the channel transit

Whether the master was fatigued could not be established, and the evidence on each side was about as likely as not to be right. In favour, the transit fell in the last hour and a quarter of a 13 hour 40 minute spread on day nine of a twelve-day tour. The master had slept 6 hours 15 minutes the night before, and he said in interview that he was tired. After sighting the beacon he eased the vessel back to port rather than stopping. He believed he was just outside the edge when he was nearly 40 m west of the centreline.

Against this, he said he was neither fighting to stay awake nor slow, and the recorded performance agrees with him. The vessel held the centreline to within 3 m before the freeze. He responded to the shallow alarm in 4 seconds. His radio calls after the grounding were complete and correct at the first attempt. He decided not to refloat in about thirty seconds.

No detailed fatigue assessment was made. Even an established fatigue would not have shown an influence, because each decision that mattered is either accounted for by the absence of a night passage plan or hold point, or contradicted by the recorded data. The matter remains unresolved. The rostering of the transit is discussed under Organisational Influences.

Individual Actions

Transit continued at eight knots from the plotter freeze until the shallow alarm

It was virtually certain that the vessel continued at transit speed from the plotter freeze at 18:07:06 until the shallow alarm at 18:11:48. The recorded track, the engine recorder, the bridge audio and the master's own account all agree on this. The engine recorder shows no throttle movement until 18:11:52. The fall in speed over the ground, from 8.4 kt at the freeze to 7.1 kt at the alarm, is accounted for by drag from the shoaling water. The bridge audio records no consideration of stopping at any point.

The master said he did not consider stopping when the plotter froze. He went straight to the visual transit because, in his mind, the visual transit was the backup. At 18:10:02 he called "There's four. Bring her back." He then knew he was west of the line. He eased back to port instead of stopping, because he took himself to be just outside the edge when he was 39 m west of the centreline. The course over the ground came back only to 197 degrees, still west of the channel's 195. In his account, stopping on the edge of a bank he could not see felt worse than easing back in.

Nothing prevented a stop. There was no other traffic, the water was flat and the wind was offshore at 8 kt. The vessel's crash stop distance from 8 kt was 92 m. The vessel crossed the channel's western edge 114 seconds after the freeze, so a stop in that time would have left it in the channel. A crash stop begun at any point before about 18:11:40 would have stopped it short of the contact point. That period included the 98 seconds after the sighting, and a stop then would have left the

vessel on the margin of the bank, upright and not holed. It was therefore extremely likely that continuing at eight knots contributed to the vessel leaving the channel and to the alarm giving too little warning to stop.

No lookout was on the bridge. The deckhand had been sent to collect fares at 18:04:14, which was normal practice on the route, and nothing in OPS-3 addressed it either way. The decision to continue was made within a passage plan that had no hold point for loss of the primary fix. That gap, and the beacon that could not be seen, are discussed under Risk Controls and Local Conditions.

Heading eased five degrees to starboard without a decision to turn

It was virtually certain that the heading eased from 195 to 200 degrees over about a minute from 18:07:06 and was then held. The recorded rate of turn peaked at 6 degrees per minute at 18:07:40 and was back to zero by 18:08:50. Rudder angle followed helm demand within 1.5 degrees throughout, so the turn was steered. The master said in interview: "I eased her to starboard while I was fiddling with the plotter and then looking for four. Five degrees. I never made that decision." His safety report says the same.

The timing places most of the turn at the plotter. The bridge audio records screen taps at 18:07:26 and "frozen again" at 18:07:44. The spoken search, "Right. Where's four", began only at 18:08:31. This was a turn that went unnoticed, not a turn made in looking for the light. Nothing on the bridge showed the master the new heading. The plotter's vessel symbol was frozen and the beacon was screened by the mangroves on Merrandi Islet.

There was no current and no wind effect at that state of tide. Steering and propulsion failure was excluded, as was a tidal set below prediction. The westing that took the vessel out of the dredged channel was therefore this heading change and nothing else. Held 5 degrees west of the channel's course, it moved the vessel west at about 20 m a minute and across the 20 m to the western edge by about 18:09:00. It was virtually certain that this contributed to the vessel leaving the channel. Whether the boat ramp floodlights delayed the moment the held heading was noticed was examined under Local Conditions and remains unresolved.

Engines stopped and no attempt made to refloat on a falling tide

It was virtually certain that both engines were stopped 27 seconds after contact and were not restarted. The pan pan at 18:13:04 included the words "I am not attempting to refloat", stated as a decision already made. The master had the No. 2 starboard void sounded. He reported 75 mm of ingress to Port Control at 18:17:05 and had the void sounded ten minutes after that. Passengers were addressed 36 seconds after the impact and again at 18:14:52, and all 38 were issued lifejackets.

The survey later found a 180 mm split in the starboard hull plating in way of that void. Ingress reached approximately 260 mm by the time the vessel refloated at 00:20. The master's reasoning took in the falling tide, the unknown bottom, the 38 people aboard, and a vessel that could not sink while it sat on the bank. Backing off would have put a flooding compartment into 3.2 m of water, with the damage unassessed and the passengers still aboard. It was very likely that the decision not to refloat reduced the risk to the passengers.

Plotter fault passed on verbally at the change of master and not entered in the defect book

It was virtually certain that the morning freezes at 10:42 and 11:17 were passed on verbally at the handover at about 13:05 and not entered in the defect book. The relieving master said the plotter

had done the same to him the month before. Both accounts agree, and both masters volunteered the omission before they were asked about it.

It was unlikely that this omission influenced the occurrence. The freeze was a firmware fault. An entry at 13:05 would have been reviewed by the Operations Manager at the end of the week. Nothing written that afternoon could have updated the firmware or taken the vessel off TF18 by 18:07. It was likely that the practice increased risk all the same. A cleared fault on the primary means of position fixing that is passed on by word of mouth leaves the next master informed and nobody ashore. This is an instance of the defect procedure gap discussed under Risk Controls, not a matter for the two masters.

Occurrence Events

Vessel left the dredged channel on the bank side

The vessel left the dredged channel on its western side, and this was virtually certain. The AIS track placed the vessel 1 m west of the charted centreline abeam the No. 2 beacon at 18:05:30. It stayed within 3 m of the centreline until 18:07:06. From that time the heading increased steadily from 195 to 200 degrees over about a minute and was then held. The vessel set west at about 20 m per minute and reached the channel's western edge at about 18:09:00, 0.40 nm north of the No. 4 beacon. By 18:10:02 it was 39 m west of the centreline. It was over Boolarong Bank from about 18:11:00. The track agrees with the surveyed grounding position and with the marks in the bed. That position was 74 m west of the centreline and 54 m outside the western edge.

The shape of the track showed a heading that was applied and then held, not a set from current or wind. The recorded rate of turn peaked at 6 degrees per minute at 18:07:40 and returned to zero by 18:08:50. The wind was offshore at 8 kt, and there was no cross-set in the channel at that state of tide. The reasons for the heading change are discussed under Individual Actions.

The master sighted the No. 4 beacon at 18:10:02. After that, the course over the ground eased back from 200 to 197 degrees. This correction slowed the westing but did not stop it, because the course stayed west of the channel's 195 degrees until contact.

The channel is 40 m wide and dredged to 3.2 m. The bank on its western side dries in places. Once the vessel was outside the western edge, the water beneath it shoaled at about 1 in 70. The grounding followed directly from the vessel being outside the dredged channel. It was virtually certain that the departure from the channel contributed to the grounding.

Shallow alarm at its 2019 setting of 0.9 m gave sixteen seconds' warning

The echo sounder's shallow alarm was set to 0.9 m under the keel. The alarm log states that this was "the value written into the setup when the unit was commissioned in 2019 and it has not been changed." The alarm sounded once on the voyage, at 18:11:48. The bridge audio records the tone beginning in that same second. Both throttles were moved to astern together 4 seconds later, and contact followed at 18:12:04. The alarm log, the bridge audio and the recorded speed agree to the second. It was virtually certain that the alarm gave 16 seconds' warning.

In those 16 seconds the vessel covered 58 m at 7.1 kt over the ground. The vessel's own trials figure for a crash stop from 8 kt is 92 m. Astern thrust was developing at the moment of contact but had not reached full. The warning therefore covered less distance than the vessel needed to stop.

The equipment did what it was set to do. On the surveyed gradient of about 1 in 70, the alarm could not have sounded appreciably earlier at a 0.9 m setting. A setting of about 1.6 m under the keel would have sounded about 107 m short of the contact. That distance would have allowed a crash stop with about four seconds to react. Such a setting still would not have sounded in the

dredged channel at low water. A setting that gave 40 seconds' warning would have sounded there. For this reason the alarm setting has to be chosen for the channel, not simply raised.

The short warning came from the 0.9 m setting combined with the speed the vessel was still making over the bank. The continued transit at speed is discussed under Individual Actions. It was extremely likely that the alarm setting contributed to the grounding, because it gave warning over a distance shorter than the vessel's stopping distance.

Steering or propulsion failure before the vessel left the centreline

The investigation considered whether a steering or propulsion failure caused the vessel to leave the centreline and found it very unlikely, because the engine monitoring recorder showed both engines steady and matched, steering pressure within its normal band and the rudder following helm demand throughout, and the whole steering system was found serviceable when removed and tested on 12/07/2026.

Findings

From the evidence available, the following findings are made with respect to the grounding of MV Tarrakin Star on Boolarong Bank in the Merrandi Channel, Kanooka Heads, on 7 July 2026. These findings should not be read as apportioning blame or liability to any particular organisation or individual.

A finding stated without a likelihood term was established as fact.

Contributing factors

- It is virtually certain that the last service was moved into darkness from 1 June 2026 without reopening the daylight-only risk assessment RRA-07 or obtaining the General Manager Operations' approval that RRA-07 required, and likely that this contributed by leaving the night transit without the specific passage plan RRA-07 called for, although a reopened assessment might have produced a night plan with no hold point in it. (Safety Issue)
- No night passage plan existed for the Merrandi Channel, and the daylight plan named no hold point or action for loss of the chart plotter as the primary means of position fixing, which is virtually certain, and it is very likely that this absence contributed by leaving nothing to prompt the master to stop when the plotter froze, when a stop begun before about 18:11:40 would still have halted the vessel short of the contact point. (Safety Issue)
- With virtual certainty, the defect procedure in OPS-9 never required a fault that cleared itself to be recorded, and this likely contributed because none of the plotter freezes of the previous year reached the Operations Manager's review, the maintenance system or the electronics contractor, so the firmware that addressed cause code 0x51 was not applied, although how quickly a recorded freeze would have led to that fix was not established, given that the operator held no subscription to the manufacturer's bulletins. (Safety Issue)
- The plotter's event log, the bridge audio and the master's account make it virtually certain that the chart plotter display froze during the channel transit at 18:07:06 and did not recover, and very likely that this contributed by removing the primary means of position fixing at the moment the heading change began.
- That the transit continued at eight knots from the plotter freeze until the shallow alarm, with no means of fixing the vessel's position, is virtually certain, and it very likely contributed, because a stop begun at any time before about 18:11:40 would have halted the vessel short of the contact point, including during the 98 seconds after the master saw the No. 4 beacon dead ahead and knew he was west of the line.
- It is virtually certain that the Kanooka Heads Port Authority deferred clearing of the No. 4 beacon's sightline, work order WO-118, as unfunded at the 2023, 2024 and 2025 works rounds, with Merrandi Islet lying outside the 50 m vegetation scope of the services agreement, and very likely that the deferral contributed, since a trim funded in 2023 would have left the mangrove canopy about 0.7 m below the lantern on the night rather than 1.6 m above it. (Safety Issue)
- The No. 4 beacon was screened by mangroves from the channel between 0.84 and 0.34 nm north of it, which is virtually certain, and this likely contributed by hiding the one remaining reference that would have shown the vessel drawing west after the heading change, although the turn itself began at the frozen plotter before the search for the beacon, and the light was clear of the islet for at least the last 37 seconds before the master sighted it.

- It is virtually certain that the master eased the heading five degrees to starboard, from 195 to 200 degrees, without a decision to turn, and virtually certain that this held heading change contributed by carrying the vessel west at about 20 m a minute, as there was no current or wind effect at that state of tide.
- The recorded track, the surveyed grounding position and the marks on the bed make it virtually certain that the vessel left the dredged channel on the bank side at about 18:09, and virtually certain that this contributed, since the vessel grounded 54 m outside the channel's western edge.
- Although the echo sounder worked exactly as set, it is virtually certain that the shallow alarm at its 2019 setting of 0.9 m gave sixteen seconds' warning, and very likely that this contributed, because at 7.1 kt the vessel covered 58 m in that time against a trials crash stop distance of 92 m.

Other factors that increased risk

- It is virtually certain that the aids code set no notice or availability duty for beacons off declared channels, because WAN-2 section 4.2 applies only to aids on a declared shipping channel and the Merrandi Channel is not one, and likely that this increased risk by leaving the users of a channel carrying a 96-passenger ferry in darkness with no means of being told of a change to its aids. (Safety Issue)
- In November 2025 the No. 4 beacon's nominal range was halved from 6 to 3 nautical miles by its solar conversion, and it is virtually certain that no notice was given to channel users, with the chart correction still pending on 7 July 2026, and likely that this increased risk because the operator's passage plan still described the beacon as a 6 M light. (Safety Issue)
- That the TF18 darkness transit was rostered to close a split duty within 20 minutes of the operator's 14 hour spread limit, on every day of a twelve-day tour and with no relief master rostered, is virtually certain, and it is likely that this increased risk on every winter darkness transit, although whether the master was fatigued on the night could not be established. (Safety Issue)
- It is virtually certain that the forward saloon lifejacket locker was blocked by a kiosk counter fitted in a 2019 refit without re-survey, so that the locker face could not be reached from 22 of the 34 forward saloon seats except through a 0.6 m gap behind the counter, and very likely that this increased risk in any emergency requiring passengers to equip themselves, with the operator advising that the same counter is fitted to its two other 24 m vessels. (Safety Issue)
- The two plotter freezes of the morning of 7 July 2026 were, with virtual certainty, passed on verbally at the change of master at about 13:05 and not entered in the defect book, and it is likely that this practice, which both masters followed and described in the same terms, increased risk by leaving a fault on the primary means of position fixing known to the next master but to nobody ashore.

Other findings

- A tide set down below prediction over Boolarong Bank was examined and ruled out, as the Port Authority's gauge read 0.91 m at 18:12 against a predicted 0.94 m, and at the predicted height the vessel would still have touched.
- A steering or propulsion failure before the vessel left the centreline was examined and ruled out on the recorded engine, steering pressure and rudder data, and by a function

test of the whole steering system on 12 July 2026 that found every component serviceable.

- The master stopped both engines 27 seconds after contact and made no attempt to refloat on a falling tide, which reduced risk because the hull was breached in way of the No. 2 starboard void and refloating would have put a flooding compartment into 3.2 m of water with the passengers still aboard.

Safety issues and actions

Clearing of the beacon's sightline deferred unfunded for three works rounds

The Kanooka Heads Port Authority deferred work order WO-118, which called for mangrove regrowth on Merrandi Islet to be cleared from the No. 4 beacon's sightline, at the 2023, 2024 and 2025 works rounds. Each entry read "defer, unfunded, refer next round". The islet lies 240 m from the beacon. That placed it outside the 50 m vegetation scope at schedule 2.4 of the aids to navigation services agreement, so the work competed for the general waterways vote rather than the aids to navigation vote. By March 2026, work order WO-176 recorded the sightline as "now fully closed on the northern approach".

It was very likely that the deferral contributed to the grounding. The canopy grew from 2.1 m in 2019 to 5.8 m in July 2026, against a lantern height of 4.2 m. A trim to 2 m funded at the 2023 round would have left the canopy about 0.7 m below the lantern on the night, rather than 1.6 m above it. The beacon would then have been open to the channel over the stretch where it was screened.

No safety action has been recorded in response to this safety issue.

Last service moved into darkness without reopening the daylight-only risk assessment

The operator moved its last Merrandi Channel service, TF18, into darkness from 1 June 2026 without reopening RRA-07, the route risk assessment that restricted the channel to daylight transits. The change, CHG-2026-014, was assessed as a timetable change only. The Operations Manager approved it on 22 April 2026. RRA-07 required any night transit to have the General Manager Operations' approval and a specific passage plan. Neither was obtained.

The change process asked whether the change introduced anything new, not whether the existing assessment still held. A restriction that named both an approver and a required control was therefore never read against the change. A 96-passenger vessel then made 31 transits of a 40 m channel in darkness. No plan covered the unlit reach between the No. 2 and No. 4 beacons, which RRA-07 already named as a hazard.

It is likely that this contributed to the occurrence by leaving the night transit without the passage plan RRA-07 called for. A night plan written for this channel was the document most likely to say what to do when the primary means of position fixing was lost. A reopened assessment might still have produced a night plan with no hold point in it. Why the change was assessed as a timetable change only was not established.

No safety action has been recorded in response to this safety issue.

No night passage plan or hold point for loss of the primary fix

Tarrakin Ferries had no night passage plan for the Merrandi Channel. Its daylight plan named no hold point or action for loss of the chart plotter, which was the primary means of position fixing. Appendix A named the visual transit of the No. 4 beacon and the town wharf leading lights as the secondary means. However, the leading lights lie about 1.9 nm beyond the No. 6 beacon and do not resolve as two lights until a vessel is inside the channel's southern third. Where the plotter was lost on the night, the secondary means was the beacon alone. OPS-3 3.6 called for reduced speed and consideration of anchoring, but only in reduced visibility, which did not apply.

A hold point was the operator's own defence against this sequence of events, and it was never written. On a flat night with no other traffic, taking the way off the vessel would have cost nothing. Both masters interviewed said independently that stopping was the right response to losing the plotter. Both also said nothing in the operator's procedures asked for it.

Safety recommendation

The investigation recommends that Tarrakin Ferries write a night passage plan for the Merrandi Channel before any further darkness transit. The plan should name a hold point: if the primary means of position fixing is lost inside the channel, the way is to be taken off the vessel until the position is re-established by another means.

[NOT ESTABLISHED: Tarrakin Ferries' response to this recommendation, and any safety action taken by the operator on this issue]

Defect procedure never requires a fault that cleared itself to be recorded

It is virtually certain that the Tarrakin Ferries Operations Manual OPS-9 (revision 4, 08/2024) contained no provision requiring a fault that had cleared itself to be recorded. Section 9.1 required immediate reporting only of a defect that made the vessel unseaworthy or affected a statutory item. The chart plotter was not a statutory item on this vessel on this route. Section 9.2 placed all other defects in the vessel defect book at the end of the shift in which they were found. That gave no trigger for a fault that had come and gone by then. Section 9.3 required the outgoing master to brief the relieving master on any defect. It did not require the brief to be written down, and it required nothing of the relieving master afterwards. Both masters independently described the same reading of the manual: a fault present at handover was written in the book, and a fault that had cleared was passed on verbally.

The issue mattered because the plotter was the primary means of position fixing named in the passage plan. None of its freezes in the year before the occurrence was written down. As a result, none reached the Operations Manager's weekly review, the maintenance system or the electronics contractor. The procedure likely contributed to the grounding, because the firmware that addressed cause code 0x51 was never applied. How quickly a recorded freeze would have led to that fix was not established, as the operator held no subscription to the manufacturer's bulletins. The procedure accounts for why the two freezes of the morning of 7 July 2026 were passed on verbally at about 13:05 and not entered in the defect book.

No safety action has been recorded in response to this issue.

Aids code set no notice or availability duty for beacons off declared channels

The Waterways Aids to Navigation Code WAN-2 set no notice or availability duty for beacons off a declared shipping channel, and it is virtually certain that this gap existed. Section 4.2 sets three duties: an availability target of 99.0 per cent, a duty to record availability, and a duty to notify any change of character, position or range by notice to mariners. These apply only to aids on a declared shipping channel, and the Merrandi Channel is not one. Its aids fall instead under section 4.3. That section requires only that the responsible authority can demonstrate an inspection interval it has set itself. Neither section 4.3 nor the services agreement that sets the interval requires availability to be recorded. Neither requires any user of the channel to be told of a defect, a deferral or a change.

It is unlikely that the gap influenced the grounding. A notice of the range change would have described a light that was screened where the vessel left the channel, and the light was lit throughout. The gap still matters. On a channel carrying a 96-passenger ferry in darkness, the operator had no right to be told when a beacon's range, character or position changed. It also had no figure for how often the aids worked. For this reason, the solar conversion of the No. 4 beacon in November 2025 reached no channel user for eight months. A future change to a light that is not screened would reach nobody either. It is likely that this increased risk. The Port Authority can close the gap through its own practice, whatever the code requires.

No safety action has been recorded in response to this safety issue.

Beacon's range halved by solar conversion with no notice to channel users

In November 2025 the No. 4 beacon was converted from a mains supply to a solar lantern under the Council's waterways energy programme. The new lantern is rated at a nominal 3 nautical miles, half the 6 of the lantern it replaced. Its character, Fl G 2.5s, did not change. On 18 July 2026 the investigator measured its luminous range from the port workboat at 2.6 nm. A notification was lodged with the hydrographic service on 4 February 2026, but the chart correction was still pending on 7 July 2026. It is virtually certain that no local notice to mariners was issued and that no operator using the channel was told. Tarrakin Ferries' current passage plan appendix, issue 2, still recorded the beacon at 6 M.

The beacon was screened by mangroves over the stretch where the vessel left the channel, and the change of range was not found to have played a part in the grounding. It is likely, however, that it increased risk. A light on a 96-passenger route had its nominal range halved, and no user of the channel was informed. The operator's passage plan described a light that was no longer there. On a night when the beacon is not screened, or where another aid is altered in the same way, a master working to that plan would be looking for a light twice as strong as the one in place.

Safety action

On 22 July 2026 local notice to waterway users LNM-2026-14 was issued, giving the No. 4 beacon's solar conversion and its 3 M nominal range. The Port Authority's aids to navigation register has been made available to every commercial operator using the channel. Tarrakin Ferries has reissued the Merrandi Channel passage plan appendix as issue 3 with the corrected range. This action is held by the Aids to Navigation Officer, Kanooka Shire Council, and is in progress.

Darkness transit rostered to close a split duty within 20 minutes of the spread limit

It is virtually certain that the winter roster placed the TF18 darkness transit of the Merrandi Channel at the end of a split duty whose spread ended 20 minutes inside the operator's 14 hour limit. This applied on every day of a 12 days on, 2 days off tour, and no relief master was rostered. The roster, the duty record and the operator's limits establish this.

Whether the master was fatigued on the night could not be established. The roster was not found to have contributed to the grounding. It is likely, however, that the roster increased risk.

The exposure is not confined to one day. It is part of the design of the winter timetable, and it recurs on every darkness transit, including those on days nine to twelve of a tour. The operator asks masters for a fatigue diary but does not read it. A master who reports unfit is covered by extending another master's tour, so no relief master is available to break a tour when that happens.

No safety action has been recorded in response to this safety issue.

Forward saloon lifejacket locker blocked by a 2019 counter fitted without re-survey

It is virtually certain that a kiosk counter fitted in a 2019 interior refit blocked access to the forward saloon lifejacket locker on MV Tarrakin Star. The refit was carried out under a minor modification notice, and that notice did not trigger a re-survey of the safety equipment arrangement. The arrangement approved at build in 2011 had an open bench in that position, with the locker face clear.

A survey of the vessel on 16 July 2026 found that the locker face could not be reached from 22 of the 34 forward saloon seats without passing behind the counter. With people standing at the counter, the only route was a 0.6 m gap on its inboard side. The muster placard directed passengers to a locker they could not see from their seats. During the muster on the night, the deckhand had to ask three passengers to move away from the counter before passing behind it to issue lifejackets.

The vessel was aground, upright and stable, and the muster took a couple of minutes longer than it should have. All 38 passengers were in lifejackets within about eight minutes. The locker arrangement had no bearing on the grounding or its outcome. It very likely increased risk in any emergency that required passengers to equip themselves without crew help, such as a fire, a capsized risk, a night evacuation into the water, or a deckhand injured in an impact. The locker remained behind the counter on this vessel. The operator advised that the same counter is fitted to its two other 24 m vessels.

No safety action has been recorded for this issue.

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